In [16…
```
#Exp :- 6
```

In [17…
```
#Aim: To perform Operation of Sampling and Population
```

In [7]:
```
#Name:Arya patil
#Roll:42
#sec:C
#subject:DSS
#Date:-07-08-2026
```

In [8]:
```
import pandas as pd
```

In [9]:
```
import os
```

In [4]:
```
os.getcwd()
```

Out[4]:
```
'C:\\Users\\aryap'
```

In [11…
```
os.chdir("C:\\Users\\aryap\\OneDrive\\Desktop")
```

In [6]:
```
data=pd.read_csv("heart.csv")
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
~\AppData\Local\Temp/ipykernel_21808/823534385.py in <module>
----> 1 data=pd.read_csv("heart.csv")

~\anaconda3\lib\site-packages\pandas\util\_decorators.py in wrapper(*args,
**kwargs)
    309                     stacklevel=stacklevel,
    310                 )
--> 311             return func(*args, **kwargs)
    312
    313         return wrapper

~\anaconda3\lib\site-packages\pandas\io\parsers\readers.py in read_csv(file
path_or_buffer, sep, delimiter, header, names, index_col, usecols, squeeze,
prefix, mangle_dupe_cols, dtype, engine, converters, true_values, false_val
ues, skipinitialspace, skiprows, skipfooter, nrows, na_values, keep_defaul
t_na, na_filter, verbose, skip_blank_lines, parse_dates, infer_datetime_for
mat, keep_date_col, date_parser, dayfirst, cache_dates, iterator, chunksiz
e, compression, thousands, decimal, lineterminator, quotechar, quoting, dou
blequote, escapechar, comment, encoding, encoding_errors, dialect, error_ba
d_lines, warn_bad_lines, on_bad_lines, delim_whitespace, low_memory, memor
y_map, float_precision, storage_options)
    584     kwds.update(kwds_defaults)
    585
--> 586     return _read(filepath_or_buffer, kwds)
    587
    588


~\anaconda3\lib\site-packages\pandas\io\parsers\readers.py in _read(filepat
h_or_buffer, kwds)
    480
    481     # Create the parser.
--> 482     parser = TextFileReader(filepath_or_buffer, **kwds)
    483
    484     if chunksize or iterator:

~\anaconda3\lib\site-packages\pandas\io\parsers\readers.py in __init__(sel
f, f, engine, **kwds)
    809             self.options["has_index_names"] = kwds["has_index_name
s"]
    810
--> 811         self._engine = self._make_engine(self.engine)
    812
    813     def close(self):

~\anaconda3\lib\site-packages\pandas\io\parsers\readers.py in _make_engin
e(self, engine)
   1038             )
   1039         # error: Too many arguments for "ParserBase"
-> 1040         return mapping[engine](self.f, **self.options)  # type: ign
ore[call-arg]
   1041
   1042     def _failover_to_python(self):
```

```
~\anaconda3\lib\site-packages\pandas\io\parsers\c_parser_wrapper.py in __in
it__(self, src, **kwds)
     49
     50         # open handles
---> 51         self._open_handles(src, kwds)
     52         assert self.handles is not None
     53

~\anaconda3\lib\site-packages\pandas\io\parsers\base_parser.py in _open_han
dles(self, src, kwds)
    220         Let the readers open IOHandles after they are done with the
ir potential raises.
    221         """
--> 222         self.handles = get_handle(
    223             src,
    224             "r",

~\anaconda3\lib\site-packages\pandas\io\common.py in get_handle(path_or_bu
f, mode, encoding, compression, memory_map, is_text, errors, storage_option
s)
    700         if ioargs.encoding and "b" not in ioargs.mode:
    701             # Encoding
--> 702             handle = open(
    703                 handle,
    704                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'heart.csv'
```

In [24…
```
data.head()
```

Out[24]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	targ
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	

In [25…
```
data.tail()
```

Out[25]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	ta
298	57	0	0	140	241	0	1	123	1	0.2	1	0	3	
299	45	1	3	110	264	0	1	132	0	1.2	1	0	3	
300	68	1	0	144	193	1	1	141	0	3.4	1	2	3	
301	57	1	0	130	131	0	1	115	1	1.2	1	1	3	
302	57	0	1	130	236	0	0	174	0	0.0	1	1	2	

In [26…
```
data.describe()
```

Out[26]:

	age	sex	cp	trestbps	chol	fbs	restecg	
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	3
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.148515	0.528053	1
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.356198	0.525860	
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000000	
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.000000	0.000000	1
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.000000	1.000000	1
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.000000	1.000000	1
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000000	2

In [27…
```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       303 non-null    int64  
 1   sex       303 non-null    int64  
 2   cp        303 non-null    int64  
 3   trestbps  303 non-null    int64  
 4   chol      303 non-null    int64  
 5   fbs       303 non-null    int64  
 6   restecg   303 non-null    int64  
 7   thalach   303 non-null    int64  
 8   exang     303 non-null    int64  
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    int64  
 11  ca        303 non-null    int64  
 12  thal      303 non-null    int64  
 13  target    303 non-null    int64  
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [28…
```
data.columns
```

Out[28]:
```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', 'thala
ch',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='str')
```

In [29…
```
list(data.columns)
```

Out[29]:
```
['age',
 'sex',
 'cp',
 'trestbps',
 'chol',
 'fbs',
 'restecg',
 'thalach',
 'exang',
 'oldpeak',
 'slope',
 'ca',
 'thal',
 'target']
```

In [30…
```
age_above_60=data[data["age"]>60]
print(age_above_60)
```

```
     age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak
\
0     63    1   3       145   233    1        0      150      0      2.3
13    64    1   3       110   211    0        0      144      1      1.8
17    66    0   3       150   226    0        1      114      0      2.6
19    69    0   3       140   239    0        1      151      0      1.8
23    61    1   2       150   243    1        1      137      1      1.0
..   ...  ...  ..       ...   ...  ...      ...      ...    ...      ...
290   61    1   0       148   203    0        1      161      0      0.0
293   67    1   2       152   212    0        0      150      0      0.8
295   63    1   0       140   187    0        0      144      1      4.0
296   63    0   0       124   197    0        1      136      1      0.0
300   68    1   0       144   193    1        1      141      0      3.4

     slope  ca  thal  target
0        0   0     1       1
13       1   0     2       1
17       0   0     2       1
19       2   2     2       1
23       1   0     2       1
..     ...  ..   ...     ...
290      2   1     3       0
293      1   0     3       0
295      2   2     3       0
296      1   0     2       0
300      1   2     3       0

[79 rows x 14 columns]
```

In [31…
```
print(len(age_above_60))
```

```
79
```

In [ ]:

In [ ]: